CRDTraj: Context and Physics Consistent Multi-Agent Trajectory Generation with Conditional-Reward Diffusion

Author Name
Affiliation
email@institution.edu

Abstract

We address the problem of generating physically plausible multi-agent trajectories conditioned on free-form textual scene descriptions, where no ground-truth trajectories exist for novel scenarios. Existing trajectory diffusion models are trained on observed data and conditioned on past motion, making them unable to generate crowd behavior for unobserved scenarios described in natural language. Meanwhile, reward-guided diffusion methods from robotics and autonomous driving assume structured environments and hard constraints that do not hold for open-ended crowd scenarios with soft behavioral expectations. We propose **CRDTraj**, a conditional-reward diffusion model that generates multi-agent trajectories consistent with both physical plausibility and contextual behavioral expectations. CRDTraj introduces three key contributions: (1) a tokenized transformer architecture that natively conditions on maps, text descriptions, and initial agent states; (2) a dual-head design with a denoising head and a multi-component reward head that provides per-component context-consistency gradients during generation; and (3) a learned adaptive guidance controller that discovers context-dependent, noise-level-aware schedules for applying reward signals during the denoising process. We train on a mixture of real pedestrian data with simple context annotations and a new synthetic benchmark of diverse context-conditioned scenarios. We introduce *Context-Specification Compliance* (CSC), a suite of metrics for evaluating context-conditioned trajectory generation without ground-truth trajectories, and demonstrate that CRDTraj outperforms simulation baselines, conditional generative models, and ablated variants across both realism and context consistency metrics.

1 Introduction

Generating realistic multi-agent trajectories for specified scenarios is a fundamental problem in crowd simulation, urban planning, safety analysis, and training data synthesis for autonomous systems. Given a spatial map M, a natural language context description C (e.g., "emergency evacuation from the east exit" or "Christmas market with browsing pedestrians"), and initial agent states S_0, the goal is to produce a set of N agent trajectories that are simultaneously *physically plausible* (respecting obstacles, body radii, smooth dynamics) and *context-consistent* (exhibiting behavioral patterns appropriate to the described scenario).

This problem differs fundamentally from trajectory *prediction*, which conditions on observed past trajectories and predicts future motion. In our setting, no past trajectories exist—we generate complete trajectories from scratch for scenarios that may never have been observed. It also differs from simulation-based approaches (e.g., Social Force Models), which produce deterministic, single-mode outputs governed by hand-tuned parameters and cannot capture the full distribution of context-appropriate human behavior.

We identify three structural gaps that prevent direct application of existing methods:

Gap 1: No conditioning interface from language to trajectory diffusion. Existing trajectory diffusion models [1, 2, 3] accept past coordinate sequences as input. None accept textual scene descriptions. The mapping from semantic context ("fire at the east exit") to behavioral distributions (fast outward movement, elevated collision tolerance, exit-seeking) is absent from the literature.

Gap 2: No reward formulation for context-dependent crowd behavior. Reward-guided diffusion methods from autonomous driving [4] encode hard physical constraints (lane-keeping, collision avoidance). Crowd behavior under diverse contexts involves *soft, scenario-dependent preferences*—the same trajectory quality that is "good" during an evacuation is "bad" at a festival. These context-conditional rewards do not exist in prior work.

Gap 3: Different reward components become evaluable at different noise levels during denoising. Speed profiles and directional flow can be estimated from coarse, noisy trajectory estimates early in denoising. Collision rates and goal achievement require near-clean trajectories and are only reliable late. A single scalar guidance weight cannot handle this heterogeneity.

We propose **CRDTraj** (**C**onditional-**R**eward **D**iffusion for **Traj**ectory generation), which addresses all three gaps with a unified architecture trained in a single stage on mixed real and synthetic data.

2 Related Work

Trajectory prediction with diffusion models. MID [1] first applied DDPM to pedestrian trajectory prediction. LED [2] introduced a leapfrog initializer for accelerated inference. Singular-Trajectory [3] unified multiple prediction tasks via SVD-based trajectory representations. These methods condition on observed past trajectories and predict future motion, operating in a fundamentally different setting from context-conditioned generation.

Reward-guided diffusion. Diffuser [4] introduced classifier-guided sampling with learned value functions for trajectory planning in RL. CTG/CTG++ [5] applied signal temporal logic constraints during diffusion sampling for traffic scenario generation. DDPO [6] and DPO for Diffusion [7] fine-tune diffusion models with reward signals without ground-truth labels. These methods operate in domains with structured constraints and existing large-scale datasets, neither of which hold for open-ended crowd scenarios.

Crowd simulation. Social Force Models (SFM) [8] produce physics-grounded trajectories but are limited by single-mode outputs, poor behavioral realism in complex scenarios, and the need for manual parameter tuning per context. Recent work has explored learning-based crowd simulation, but none condition on free-form text descriptions.

Language-conditioned motion. LMTrajectory [9] uses language models for trajectory prediction but operates in the prediction setting with observed history. Trace and Pace [10] uses guided diffusion for controllable pedestrian animation but requires structured goal specifications rather than free-form text.

3 Problem Formulation

3.1 Task Definition

Given:

- A spatial map $M \in \mathbb{R}^{H \times W}$ (obstacle/walkable region image),
- A natural language context description C (e.g., "emergency evacuation toward south exits"),
- Initial agent states $S_0 = \{s_0^i\}_{i=1}^N$ where $s_0^i = (x_0^i, y_0^i, v_{x,0}^i, v_{y,0}^i, g_x^i, g_y^i) \in \mathbb{R}^6$ encodes position, velocity, and goal,
- A context-derived behavioral specification $\mathcal{B}(C)$ consisting of soft target ranges for observable behavioral metrics,

generate a set of N agent trajectories $\boldsymbol{\tau} = \{\boldsymbol{\tau}^i\}_{i=1}^N$ where $\boldsymbol{\tau}^i = [(x_1^i, y_1^i), \ldots, (x_T^i, y_T^i)] \in \mathbb{R}^{T \times 2}$, such that $\boldsymbol{\tau}$ is:

1. **Physically plausible**: trajectories respect map obstacles, exhibit smooth dynamics, and maintain reasonable inter-agent spacing.
2. **Context-consistent**: observable behavioral metrics of $\boldsymbol{\tau}$ fall within the specification $\mathcal{B}(C)$.
3. **Diverse**: multiple samples from the model produce meaningfully different yet valid trajectories.

3.2 Context-Derived Behavioral Specification

The behavioral specification $\mathcal{B}(C)$ is generated by a large language model from the textual context C and consists of:

$$\begin{aligned}\mathcal{B}(C) = \big\{&[v_{\min}, v_{\max}],\ [c_{\min}^{\text{early}}, c_{\max}^{\text{early}}],\ [c_{\min}^{\text{late}}, c_{\max}^{\text{late}}],\\ &d_{\text{event}} \in \{\text{toward}, \text{away}\},\ d_{\text{goal}} \in \{\text{true}, \text{false}\},\\ &[\rho_{\min}, \rho_{\max}],\ [\ell_{\min}, \ell_{\max}]\big\}\end{aligned} \tag{1}$$

where $[v_{\min}, v_{\max}]$ is the desired speed range, $[c_{\min}^{\text{early/late}}, c_{\max}^{\text{early/late}}]$ are early/late collision rate ranges, d_{event} and d_{goal} are boolean directionality fields, $[\rho_{\min}, \rho_{\max}]$ is the relevance-to-event range, and $[\ell_{\min}, \ell_{\max}]$ is the lingering fraction range.

4 CRDTraj Architecture

CRDTraj is a conditional denoising diffusion model with a tokenized transformer backbone and dual output heads: a **denoising head** for noise estimation and a **multi-component reward head** for context-consistency scoring. An **adaptive guidance controller** modulates per-component reward influence at each denoising step.

4.1 Diffusion Process

We adopt the standard DDPM framework. Let $\boldsymbol{\tau}_0 \in \mathbb{R}^{N \times T \times 2}$ be the clean multi-agent trajectory tensor. The forward process adds Gaussian noise over Γ steps:

$$q(\boldsymbol{\tau}_t | \boldsymbol{\tau}_0) = \mathcal{N}\big(\boldsymbol{\tau}_t;\ \sqrt{\bar{\alpha}_t}\,\boldsymbol{\tau}_0,\ (1 - \bar{\alpha}_t)\,\mathbf{I}\big) \tag{2}$$

where $\bar{\alpha}_t = \prod_{\gamma=1}^t \alpha_\gamma$ and $\{\alpha_\gamma\}$ is the noise schedule.

The reverse (denoising) process generates trajectories by iteratively removing noise:

$$p_\theta(\boldsymbol{\tau}_{t-1} | \boldsymbol{\tau}_t, M, C, S_0) = \mathcal{N}\big(\boldsymbol{\tau}_{t-1};\ \boldsymbol{\mu}_\theta(\boldsymbol{\tau}_t, t, M, C, S_0),\ \sigma_t^2 \mathbf{I}\big) \tag{3}$$

where $\boldsymbol{\mu}_\theta$ is parameterized through noise prediction $\boldsymbol{\epsilon}_\theta$:

$$\boldsymbol{\mu}_\theta(\boldsymbol{\tau}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\boldsymbol{\tau}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\boldsymbol{\tau}_t, t, M, C, S_0) \right) \tag{4}$$

At each step, the clean trajectory estimate is obtained via Tweedie's formula:

$$\hat{\boldsymbol{\tau}}_0 = \frac{\boldsymbol{\tau}_t - \sqrt{1-\bar{\alpha}_t}\, \boldsymbol{\epsilon}_\theta}{\sqrt{\bar{\alpha}_t}} \tag{5}$$

4.2 Tokenization and Input Encoding

All inputs are converted to token sequences in a shared embedding space of dimension d.

Agent tokens. Each agent i's noisy trajectory $\boldsymbol{\tau}_t^i \in \mathbb{R}^{T\times 2}$, concatenated with initial state $s_0^i \in \mathbb{R}^6$, is projected via a per-agent MLP:

$$\mathbf{h}_{\text{agent}}^i = \text{MLP}_{\text{agent}}\big([\boldsymbol{\tau}_t^i \,\|\, s_0^i]\big) \in \mathbb{R}^d \tag{6}$$

yielding N agent tokens $\{\mathbf{h}_{\text{agent}}^i\}_{i=1}^N$.

Map tokens. The map image M is processed by a lightweight CNN (e.g., ResNet-18) and spatially flattened into P patch tokens:

$$\{\mathbf{h}_{\text{map}}^p\}_{p=1}^P = \text{CNN}_{\text{map}}(M), \quad \mathbf{h}_{\text{map}}^p \in \mathbb{R}^d \tag{7}$$

Context tokens. The text description C is encoded by a frozen pretrained language model (e.g., Sentence-BERT) followed by a trainable projection MLP:

$$\{\mathbf{h}_{\text{ctx}}^l\}_{l=1}^L = \text{MLP}_{\text{ctx}}(\text{LM}_{\text{frozen}}(C)), \quad \mathbf{h}_{\text{ctx}}^l \in \mathbb{R}^d \tag{8}$$

Timestep embedding. The diffusion timestep t is encoded via sinusoidal positional encoding followed by an MLP:

$$\mathbf{h}_t = \text{MLP}_t(\text{SinEmb}(t)) \in \mathbb{R}^d \tag{9}$$

This is added to all tokens as a global conditioning signal.

Unified token sequence. The full input sequence is:

$$\mathbf{H}^{(0)} = \big[\mathbf{h}_{\text{agent}}^1, \ldots, \mathbf{h}_{\text{agent}}^N,\ \mathbf{h}_{\text{map}}^1, \ldots, \mathbf{h}_{\text{map}}^P,\ \mathbf{h}_{\text{ctx}}^1, \ldots, \mathbf{h}_{\text{ctx}}^L\big] + \mathbf{h}_t \tag{10}$$

4.3 Transformer Backbone

The backbone consists of $\mathcal{L}$ transformer blocks, each containing:

Self-attention (agent–agent interaction). Agent tokens attend to each other, learning social coordination and collision avoidance:

$$\mathbf{H}'_{\text{agent}} = \text{SelfAttn}(\mathbf{H}_{\text{agent}}) + \mathbf{H}_{\text{agent}} \tag{11}$$

Cross-attention (agent → map + context). Agent tokens query map and context tokens, learning spatial feasibility and behavioral conditioning:

$$\mathbf{H}''_{\text{agent}} = \text{CrossAttn}\big(\mathbf{H}'_{\text{agent}},\ [\mathbf{H}_{\text{map}} \,\|\, \mathbf{H}_{\text{ctx}}]\big) + \mathbf{H}'_{\text{agent}} \tag{12}$$

Feed-forward and normalization. Standard FFN with LayerNorm:

$$\mathbf{H}^{(\ell+1)} = \text{LayerNorm}\big(\text{FFN}(\mathbf{H}''_{\text{agent}}) + \mathbf{H}''_{\text{agent}}\big) \tag{13}$$

After $\mathcal{L}$ layers, we obtain contextualized agent representations $\{\mathbf{z}^i\}_{i=1}^N$.

4.4 Denoising Head

The denoising head maps each agent's representation to a noise estimate:

$$\boldsymbol{\epsilon}_\theta^i = \text{MLP}_{\text{denoise}}(\mathbf{z}^i) \in \mathbb{R}^{T\times 2} \tag{14}$$

The full noise estimate is $\boldsymbol{\epsilon}_\theta = \{\boldsymbol{\epsilon}_\theta^i\}_{i=1}^N \in \mathbb{R}^{N\times T\times 2}$.

4.5 Multi-Component Reward Head

The reward head predicts per-component context-consistency scores from the clean trajectory estimate $\hat{\boldsymbol{\tau}}_0$ (Eq. 5) and the contextualized backbone features. Let $\bar{\mathbf{z}} = \frac{1}{N}\sum_i \mathbf{z}^i$ be the pooled agent representation. The reward head outputs:

$$[\hat{r}_1, \hat{r}_2, \hat{r}_3, \hat{r}_4, \hat{r}_5, \hat{r}_6] = \text{MLP}_{\text{reward}}([\bar{\mathbf{z}} \,\|\, \text{Pool}(\hat{\boldsymbol{\tau}}_0)]) \in \mathbb{R}^6 \tag{15}$$

where $\text{Pool}(\hat{\boldsymbol{\tau}}_0)$ extracts trajectory-level statistics (mean speed, spatial spread, directional features) from the estimated clean trajectory. Each $\hat{r}_i$ estimates the corresponding sub-reward $r_i(\boldsymbol{\tau}_0)$ described in Section 5.

5 Reward Function Design

The reward function $R(\boldsymbol{\tau}, C, M)$ consists of six differentiable sub-reward components, each mapping to a field in the behavioral specification $\mathcal{B}(C)$. All sub-rewards are bounded in $[0, 1]$.

5.1 Band-Shaped Reward Template

For specification fields that define a target range $[a, b]$, we use a smooth band-shaped reward:

$$\text{Band}(x;\ a, b, \sigma) = \exp\left(-\frac{\max(0,\ a - x)^2}{2\sigma^2}\right) \cdot \exp\left(-\frac{\max(0,\ x - b)^2}{2\sigma^2}\right) \tag{16}$$

This function equals 1.0 when $x \in [a, b]$ and decays smoothly (with bandwidth σ) outside the band. Critically, there is *no gradient inside the band*—once a metric enters the acceptable range, the reward head stops pushing. Outside the band, the gradient points inward.

5.2 Sub-Reward Components

r_1**: Speed compliance.** Measures whether agent speeds match the desired range $[v_{\min}, v_{\max}]$:

$$\bar{v}^i = \frac{1}{T-1}\sum_{t=1}^{T-1} \frac{\|\text{pos}_{t+1}^i - \text{pos}_t^i\|}{\Delta t}, \qquad r_{\text{speed}}(\boldsymbol{\tau}) = \frac{1}{N}\sum_{i=1}^{N} \text{Band}(\bar{v}^i;\ v_{\min}, v_{\max}, \sigma_v) \tag{17}$$

r_2**: Early collision rate.** Measures collision frequency in the first half of the trajectory against $[c_{\min}^{\text{early}}, c_{\max}^{\text{early}}]$. We use a soft collision indicator for differentiability:

$$\text{col}_{\text{soft}}(i,j,t) = \text{sigmoid}\left(\frac{d_{\text{col}} - \|\text{pos}_t^i - \text{pos}_t^j\|}{\xi}\right) \tag{18}$$

where d_{col} is the collision distance threshold and ξ is the temperature. The early collision rate and its reward:

$$\bar{c}_{\text{early}} = \frac{2}{N(N-1)\cdot\lfloor T/2\rfloor}\sum_{t=1}^{\lfloor T/2\rfloor}\sum_{i<j}\text{col}_{\text{soft}}(i,j,t), \quad r_{\text{col-e}}(\boldsymbol{\tau}) = \text{Band}(\bar{c}_{\text{early}};\ c_{\min}^{\text{early}}, c_{\max}^{\text{early}}, \sigma_c) \tag{19}$$

r_3**: Late collision rate.** Identical structure to r_2 but computed over the second half of the trajectory ($t \in [\lfloor T/2\rfloor + 1,\, T]$) against $[c_{\min}^{\text{late}}, c_{\max}^{\text{late}}]$:

$$r_{\text{col-l}}(\boldsymbol{\tau}) = \text{Band}(\bar{c}_{\text{late}};\ c_{\min}^{\text{late}}, c_{\max}^{\text{late}}, \sigma_c) \tag{20}$$

r_4**: Event directionality.** Measures whether agents move toward or away from the event center $\mathbf{e} \in \mathbb{R}^2$, modulated by the relevance-to-event range $[\rho_{\min}, \rho_{\max}]$. Let $\Delta d^i = \|\text{pos}_T^i - \mathbf{e}\| - \|\text{pos}_0^i - \mathbf{e}\|$ (positive means moved away):

$$r_{\text{dir}}^i = \begin{cases} \text{sigmoid}(-\Delta d^i/\xi_e) & \text{if } d_{\text{event}} = \text{toward} \\ \text{sigmoid}(\Delta d^i/\xi_e) & \text{if } d_{\text{event}} = \text{away} \end{cases} \tag{21}$$

The relevance measures how strongly the directional signal manifests:

$$\rho = \frac{1}{N}\sum_i r_{\text{dir}}^i, \qquad r_{\text{event}}(\boldsymbol{\tau}) = \text{Band}(\rho;\ \rho_{\min}, \rho_{\max}, \sigma_\rho) \tag{22}$$

r_5**: Goal achievement.** When $d_{\text{goal}} = \text{true}$, measures proximity of each agent's final position to its assigned goal $\mathbf{g}^i$:

$$r_{\text{goal}}(\boldsymbol{\tau}) = \begin{cases} \frac{1}{N}\sum_{i=1}^N \exp\left(-\frac{\|\text{pos}_T^i - \mathbf{g}^i\|^2}{2\sigma_g^2}\right) & \text{if } d_{\text{goal}} = \text{true} \\ 1.0 & \text{if } d_{\text{goal}} = \text{false} \end{cases} \tag{23}$$

r_6**: Lingering fraction.** Measures the fraction of agents that exhibit slow/stationary behavior for sustained periods, against $[\ell_{\min}, \ell_{\max}]$. An agent i is considered lingering if its instantaneous speed drops below threshold v_ℓ for at least k consecutive timesteps. For differentiability:

$$\ell_{\text{soft}}^i = \max_{t_0}\prod_{t=t_0}^{t_0+k-1}\text{sigmoid}\left(\frac{v_\ell - \bar{v}_t^i}{\xi_\ell}\right), \qquad r_{\text{linger}}(\boldsymbol{\tau}) = \text{Band}\left(\frac{1}{N}\sum_i \ell_{\text{soft}}^i;\ \ell_{\min}, \ell_{\max}, \sigma_\ell\right) \tag{24}$$

5.3 Combined Reward

The total reward uses a multiplicative structure to enforce simultaneous satisfaction of all components:

$$R(\boldsymbol{\tau}, C, M) = \prod_{k=1}^{6} r_k(\boldsymbol{\tau}) \tag{25}$$

The multiplicative form ensures that if any single component is near zero, the total reward collapses, forcing the model to address the worst-performing aspect first. The gradient decomposes naturally:

$$\nabla_{\boldsymbol{\tau}} R = \sum_{k=1}^{6} \left(\prod_{j \neq k} r_j \right) \nabla_{\boldsymbol{\tau}} r_k \tag{26}$$

6 Guided Denoising with Adaptive Controller

6.1 Per-Component Reward Guidance

At each denoising step t, the reward head produces per-component estimates $[\hat{r}_1, \ldots, \hat{r}_6]$ from $\hat{\boldsymbol{\tau}}_0$. Each component has its own gradient with respect to the noisy trajectory:

$$\nabla_{\boldsymbol{\tau}_t} \hat{r}_k = \frac{\partial \hat{r}_k}{\partial \hat{\boldsymbol{\tau}}_0} \cdot \frac{\partial \hat{\boldsymbol{\tau}}_0}{\partial \boldsymbol{\tau}_t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \cdot \frac{\partial \hat{r}_k}{\partial \hat{\boldsymbol{\tau}}_0} \tag{27}$$

The guided denoising step replaces the standard update:

$$\boxed{\boldsymbol{\tau}_{t-1} = \boldsymbol{\mu}_\theta(\boldsymbol{\tau}_t, t) + \sum_{k=1}^{6} g_k(t) \cdot \nabla_{\boldsymbol{\tau}_t} \hat{r}_k + \sigma_t \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \tag{28}$$

where $g_k(t) \in [0, g_{\max}]$ is the per-component gate controlling the influence of sub-reward k at step t.

6.2 Noise-Level Reliability of Sub-Rewards

Different sub-rewards become reliable estimators at different noise levels, depending on whether they measure global/scalar features or fine-grained relational features:

Sub-reward	**Feature type**	**Reliable from**
r_{speed}	Global scalar average	Early (high noise)
r_{event}	Global directional	Early–mid
r_{linger}	Per-agent temporal	Mid
$r_{\text{col-early}}$	Pairwise spatial	Late (low noise)
$r_{\text{col-late}}$	Pairwise spatial	Late
r_{goal}	Per-agent endpoint	Very late

A single scalar guidance weight $w(t)$ cannot capture this heterogeneity: setting it high at noisy steps amplifies unreliable collision gradients, while setting it low suppresses useful speed and direction signals.

6.3 Adaptive Guidance Controller w_ψ

We introduce a small learned network w_ψ that outputs per-component gates conditioned on the current denoising state:

$$[g_1(t), \ldots, g_6(t)] = g_{\max} \cdot \text{sigmoid}\Big(\text{MLP}_\psi\Big(\text{SNR}(t),\ \mathbf{e}_{\text{ctx}},\ \hat{R}_t,\ [\|\nabla \hat{r}_k\|]_{k=1}^{6}\Big)\Big) \tag{29}$$

where:

- $\mathrm{SNR}(t) = \bar{\alpha}_t/(1-\bar{\alpha}_t)$ is the signal-to-noise ratio at step t,
- $\mathbf{e}_{\text{ctx}} \in \mathbb{R}^d$ is the context embedding (reused from the backbone),
- $\hat{R}_t = \prod_k \hat{r}_k$ is the current total reward estimate,
- $\|\nabla \hat{r}_k\|$ is the gradient norm of each sub-reward component.

The controller is a 3-layer MLP with 64 hidden units (∼50K parameters), outputting a 6-dimensional gate vector.

Controller training. The controller is trained after the main model (Stage 2), with backbone and both heads frozen. Training uses REINFORCE [13] over full denoising rollouts:

$$\mathcal{L}_{\text{ctrl}} = -\mathbb{E}_{\boldsymbol{\tau}_0 \sim p_\theta}[R(\boldsymbol{\tau}_0, C, M)] + \beta \sum_{t=1}^{\Gamma} \sum_{k=1}^{6} g_k(t)^2 \tag{30}$$

The first term maximizes the final trajectory reward. The second term (coefficient β) regularizes against excessive guidance, preventing physically implausible trajectories that override the denoising prior. The gradient is estimated via:

$$\nabla_\psi \mathcal{L}_{\text{ctrl}} \approx -\frac{1}{B} \sum_{b=1}^{B} \left(R(\boldsymbol{\tau}_0^{(b)}) - \bar{R} \right) \nabla_\psi \log \pi_\psi(\mathbf{g}^{(b)}) + 2\beta \sum_{t,k} g_k(t) \nabla_\psi g_k(t) \tag{31}$$

where $\bar{R}$ is a running average baseline for variance reduction and B is the rollout batch size.

7 Training

7.1 Data

We train on a mixture of two data sources:

Real data. ETH/UCY [11] and Stanford Drone Dataset (SDD) [12], annotated with simple context descriptions per scene (e.g., "Pedestrians crossing in front of a university building during daytime"). Multiple paraphrases per scene provide text diversity. These provide ground-truth human trajectories with simple contexts.

Synthetic data. A new benchmark of diverse context-conditioned scenarios generated via LLM-authored behavioral specifications and Social Force Model simulation. Each scenario includes a map, natural language context, initial agent states, and behavioral specification $\mathcal{B}(C)$. Scenarios span categories including evacuation, festival, aggressive encounters, normal pedestrian traffic, and protests.

Real data is oversampled to maintain a balanced ratio, anchoring the model on realistic human motion patterns.

7.2 Joint Training Loss

The model trains in a single stage with a combined loss:

$$\boxed{\mathcal{L} = \underbrace{\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\boldsymbol{\tau}_t, t, M, C, S_0)\|^2}_{\mathcal{L}_{\text{denoise}}} + \lambda \sum_{k=1}^{6} \underbrace{\lambda_k(t) \cdot \|\hat{r}_k(\hat{\boldsymbol{\tau}}_0) - r_k(\boldsymbol{\tau}_0)\|^2}_{\mathcal{L}_{\text{reward},k}}} \tag{32}$$

where:

- $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the sampled noise and $t \sim \text{Uniform}\{1, \ldots, \Gamma\}$,
- $r_k(\boldsymbol{\tau}_0)$ is the ground-truth sub-reward computed on the clean trajectory,
- $\hat{r}_k(\hat{\boldsymbol{\tau}}_0)$ is the reward head's prediction from the noisy estimate,
- $\lambda_k(t)$ is a noise-level-dependent weight that downweights unreliable components at high noise:

$$\lambda_k(t) = \text{sigmoid}\left(\frac{\text{SNR}(t) - \kappa_k}{\xi_k}\right) \tag{33}$$

where κ_k is the SNR threshold below which component k's prediction target becomes unreliable (determined on validation data), and ξ_k controls the transition sharpness.

On real data, $r_k(\boldsymbol{\tau}_0)$ reflects genuine human behavior. On synthetic data, $r_k(\boldsymbol{\tau}_{\text{SFM}})$ reflects SFM's partial achievement. The reward head learns a smooth function from trajectory features to reward that can extrapolate beyond SFM quality at inference time.

7.3 Overall Training Procedure

Algorithm 1 CRDTraj Training

Stage 1: Joint backbone + heads training
repeat
  Sample $(\boldsymbol{\tau}_0, M, C, S_0)$ from mixed real + synthetic data
  Sample $t \sim \text{Uniform}\{1, \ldots, \Gamma\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  $\boldsymbol{\tau}_t = \sqrt{\bar{\alpha}_t}\, \boldsymbol{\tau}_0 + \sqrt{1 - \bar{\alpha}_t}\, \boldsymbol{\epsilon}$
  Forward pass: compute $\boldsymbol{\epsilon}_\theta$, $\hat{\boldsymbol{\tau}}_0$ via Tweedie, $[\hat{r}_1, \ldots, \hat{r}_6]$
  Compute $r_k(\boldsymbol{\tau}_0)$ for each sub-reward on clean trajectory
  $\mathcal{L} = \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\|^2 + \lambda \sum_k \lambda_k(t)\, \|\hat{r}_k - r_k\|^2$
  Update θ (backbone + denoising head + reward head)
until convergence

Stage 2: Controller training (freeze θ)
repeat
  Sample batch of B scenarios from synthetic data
  for each scenario **do**
    Run full denoising rollout with w_ψ producing gates at each step
    Compute $R(\boldsymbol{\tau}_0^{\text{gen}}, C, M)$ on generated trajectory
  end for
  $\mathcal{L}_{\text{ctrl}} = -\frac{1}{B} \sum_b (R^{(b)} - \bar{R}) \log \pi_\psi + \beta \sum_{t,k} g_k(t)^2$
  Update ψ (controller only)
until convergence

8 Inference

Algorithm 2 CRDTraj Inference

Require: Map M, context C, initial states S_0, diffusion steps Γ

Encode: $\mathbf{e}_{\text{ctx}} = \text{MLP}_{\text{ctx}}(\text{LM}(C))$, map tokens, agent tokens from S_0
$\boldsymbol{\tau}_\Gamma \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \in \mathbb{R}^{N \times T \times 2}$
for $t = \Gamma, \ldots, 1$ **do**
  $\boldsymbol{\epsilon}_\theta,\ [\hat{r}_1, \ldots, \hat{r}_6] = \text{CRDTraj}(\boldsymbol{\tau}_t, t, M, C, S_0)$
  $\hat{\boldsymbol{\tau}}_0 = (\boldsymbol{\tau}_t - \sqrt{1 - \bar{\alpha}_t}\, \boldsymbol{\epsilon}_\theta)/\sqrt{\bar{\alpha}_t}$ ▷ Tweedie estimate
  $[g_1, \ldots, g_6] = w_\psi(\text{SNR}(t),\ \mathbf{e}_{\text{ctx}},\ \hat{R}_t,\ [\|\nabla \hat{r}_k\|])$ ▷ Adaptive gates
  $\boldsymbol{\tau}_{t-1} = \boldsymbol{\mu}_\theta(\boldsymbol{\tau}_t) + \sum_k g_k \cdot \nabla_{\boldsymbol{\tau}_t} \hat{r}_k + \sigma_t \mathbf{z}$ ▷ Guided step
end for
return $\boldsymbol{\tau}_0$

9 Evaluation

We evaluate along two independent axes: **realism** (does the trajectory look like real human motion?) and **context consistency** (does it match the behavioral specification?).

9.1 Realism Metrics (on Public Datasets)

Evaluated on ETH/UCY and SDD with held-out test sets.

Heatmap KL divergence (↓). Spatial occupancy heatmaps from generated and ground-truth trajectories (discretized into an $H' \times W'$ grid). We measure $D_{\text{KL}}(p_{\text{gen}} \| p_{\text{gt}})$.

Flow map cosine similarity (↑). Per-cell average velocity vectors for generated and ground-truth trajectories. We report the mean cosine similarity of the flow fields.

Speed distribution KL (↓). Empirical speed distributions of generated and ground-truth trajectories; measured via D_{KL}.

Discriminator score (↑). A trajectory-level binary classifier trained to distinguish real from generated trajectories on motion features (acceleration smoothness, speed distribution, turning angle distribution). We report the probability assigned to "real" for generated trajectories; 0.5 indicates indistinguishable.

Diversity (↑). For K generated trajectory sets per scenario, the average pairwise L2 distance between sets.

9.2 Context-Specification Compliance (CSC)

A new metric suite for evaluating context-conditioned trajectory generation. For each scenario with specification $\mathcal{B}(C)$, we measure whether the generated trajectories satisfy each behavioral target:

$$\mathrm{CSC}_{\text{speed}} = \frac{1}{N}\sum_i \mathbf{1}\big[\bar{v}^i \in [v_{\min}, v_{\max}]\big] \tag{34}$$

$$\mathrm{CSC}_{\text{col-e}} = \mathbf{1}\Big[\bar{c}_{\text{early}} \in [c_{\min}^{\text{early}}, c_{\max}^{\text{early}}]\Big] \tag{35}$$

$$\mathrm{CSC}_{\text{col-l}} = \mathbf{1}\Big[\bar{c}_{\text{late}} \in [c_{\min}^{\text{late}}, c_{\max}^{\text{late}}]\Big] \tag{36}$$

$$\mathrm{CSC}_{\text{event}} = \frac{1}{N}\sum_i \mathbf{1}\big[\mathrm{sign}(\Delta d^i) = \text{expected from } d_{\text{event}}\big] \tag{37}$$

$$\mathrm{CSC}_{\text{goal}} = \begin{cases} \frac{1}{N}\sum_i \mathbf{1}\big[\|\mathrm{pos}_T^i - \mathbf{g}^i\| < r_g\big] & \text{if } d_{\text{goal}} = \text{true} \\ \text{N/A} & \text{otherwise} \end{cases} \tag{38}$$

$$\mathrm{CSC}_{\text{linger}} = \mathbf{1}\big[\bar{\ell} \in [\ell_{\min}, \ell_{\max}]\big] \tag{39}$$

Overall CSC (↑). The fraction of scenarios where *all applicable* specifications are simultaneously satisfied:

$$\mathrm{CSC}_{\text{all}} = \frac{1}{|\mathcal{D}_{\text{test}}|}\sum_{s\in\mathcal{D}_{\text{test}}} \prod_{k\in\text{applicable}(s)} \mathrm{CSC}_k(s) \tag{40}$$

9.3 Baselines

- **SFM (oracle)**: Social Force Model with specification fields fed directly as simulation parameters.
- **Conditional VAE**: CVAE trained on the same mixed data with context conditioning.
- **Conditional GAN**: CGAN with context-conditioned generator and discriminator.
- **CRDTraj (no reward)**: Ablation—pure conditional diffusion, reward head removed.
- **CRDTraj (single w, fixed)**: Single scalar guidance weight, hand-tuned ramp.
- **CRDTraj (single w, learned)**: Learned scalar $w_\psi(t, \mathbf{e}_{\text{ctx}})$, no per-component split.
- **CRDTraj (per-comp., fixed gates)**: Per-component gates from SNR thresholds.
- **CRDTraj (full)**: Full model with learned per-component adaptive controller.

9.4 Ablation Tables

Table 1: Realism evaluation on ETH/UCY and SDD.

Method	**Heat. KL↓**	**Flow Cos↑**	**Speed KL↓**	**Discrim.↑**	**Diversity↑**
SFM (oracle)	–	–	–	–	–
Cond. VAE	–	–	–	–	–
Cond. GAN	–	–	–	–	–
CRDTraj (no reward)	–	–	–	–	–
CRDTraj (full)	–	–	–	–	–

Table 1: Realism (public datasets).

Table 2: Context consistency (synthetic benchmark).

Table 2: Context-Specification Compliance on synthetic benchmark.

Method	**Speed**	**Col-E**	**Col-L**	**Event**	**Linger**	**CSC$_{\text{all}}$**
SFM (oracle)	–	–	–	–	–	–
Cond. VAE	–	–	–	–	–	–
Cond. GAN	–	–	–	–	–	–
CRDTraj (no reward)	–	–	–	–	–	–
CRDTraj (single w, fixed)	–	–	–	–	–	–
CRDTraj (single w, learned)	–	–	–	–	–	–
CRDTraj (per-comp., fixed)	–	–	–	–	–	–
CRDTraj (full)	–	–	–	–	–	–

9.5 Expected Findings

1. Per-component learned gates outperform all alternatives, with the largest gains on collision and goal metrics that require late-step precision.
2. The learned controller discovers context-dependent schedules: evacuation contexts activate goal guidance earlier; market contexts sustain lingering guidance; aggressive contexts elevate early collision tolerance.
3. CRDTraj (full) matches or exceeds SFM (oracle) on CSC while producing substantially more realistic and diverse trajectories.
4. On public datasets, CRDTraj achieves competitive realism metrics despite being designed for a harder task (generation, not prediction).

10 Contributions Summary

1. **New task and benchmark.** We formalize context-conditioned multi-agent trajectory generation and introduce a synthetic benchmark with LLM-authored behavioral specifications spanning diverse scenarios.

2. **CRDTraj architecture.** A purpose-built conditional-reward diffusion model with tokenized inputs, dual denoising/reward heads, and per-component reward gradients, trained in a single stage on mixed real and synthetic data.

3. **Adaptive guidance controller.** A learned per-component gating mechanism that discovers noise-level-aware, context-dependent guidance schedules, addressing the fundamental challenge that different behavioral metrics become evaluable at different stages of the denoising process.

4. **Context-Specification Compliance (CSC).** A new evaluation framework for context-conditioned generation that directly measures compliance with behavioral specifications without requiring ground-truth trajectories.

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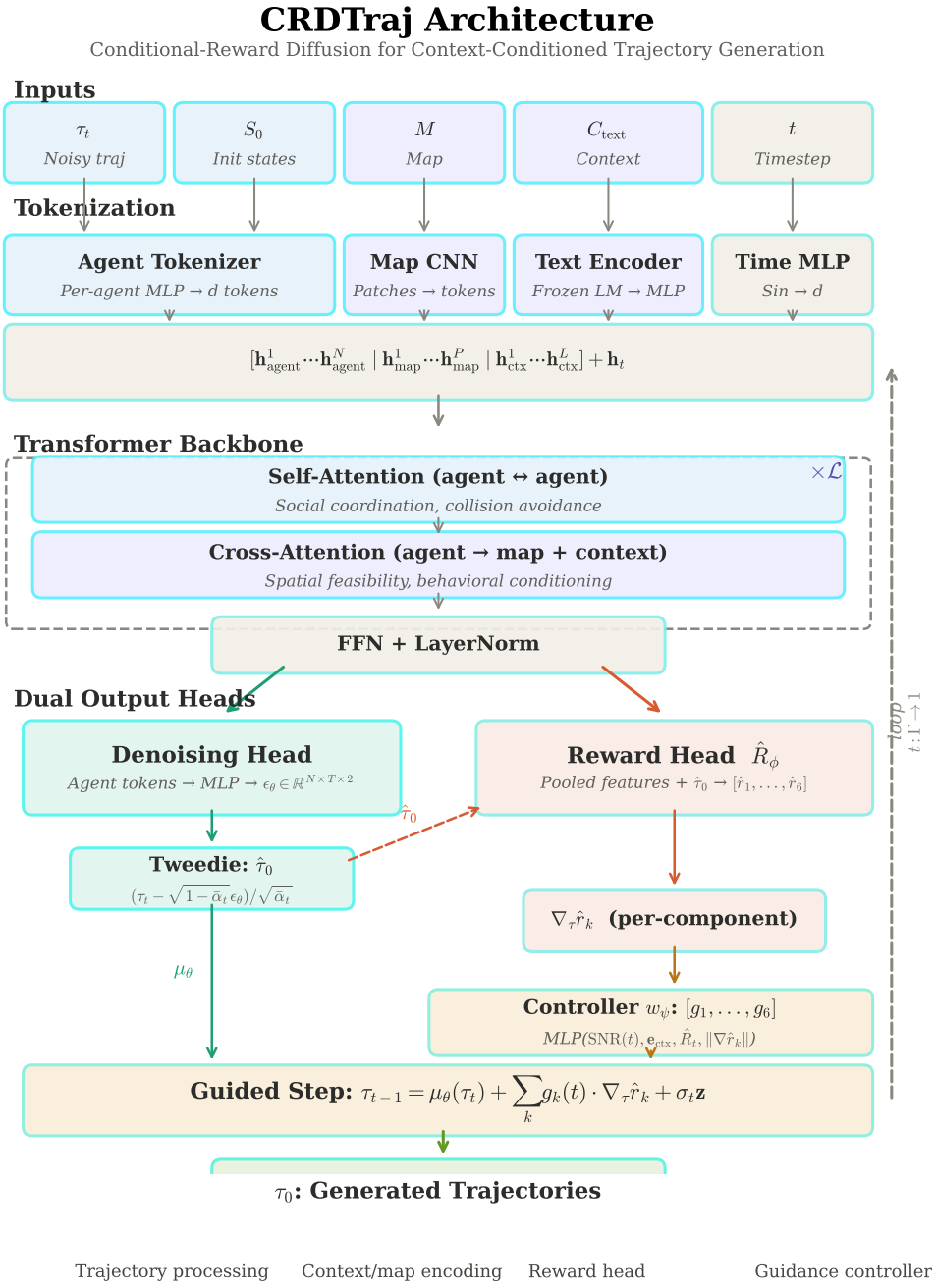

Figure 1: **CRDTraj architecture.** Inputs (noisy trajectory $\boldsymbol{\tau}_t$, initial states S_0, map M, context C, timestep t) are tokenized into a unified sequence. A transformer backbone with self-attention (agent–agent) and cross-attention (agent–map/context) produces contextualized representations. The denoising head predicts noise $\boldsymbol{\epsilon}_\theta$, from which the clean trajectory estimate $\hat{\boldsymbol{\tau}}_0$ is obtained via Tweedie's formula. The reward head evaluates $\hat{\boldsymbol{\tau}}_0$ for per-component context consistency $[\hat{r}_1, \ldots, \hat{r}_6]$. The adaptive guidance controller w_ψ produces per-component gates $[g_1, \ldots, g_6]$ that modulate the reward gradients in the guided denoising step. The loop repeats for Γ steps.

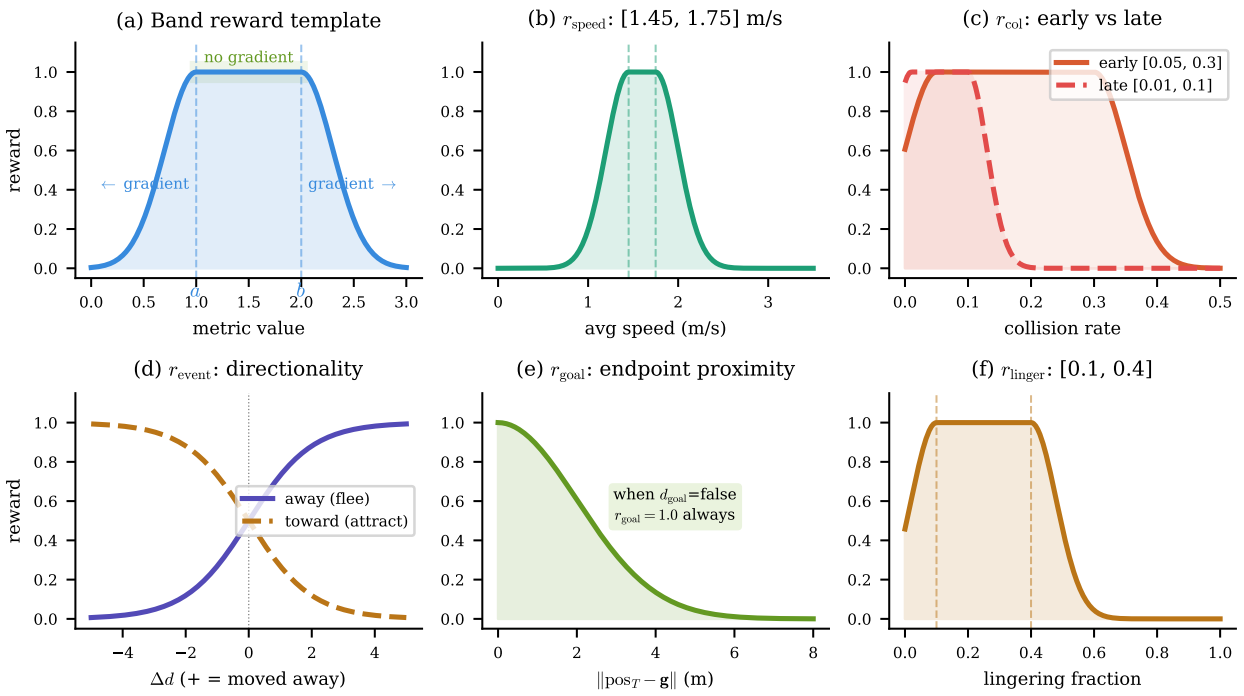

Figure 2: **Sub-reward components.** (a) The band-shaped reward template gives full reward inside the target range $[a, b]$ with zero gradient, and smooth Gaussian falloff outside with inward-pointing gradients. (b)–(f) Each sub-reward uses the band template with context-specific ranges from the behavioral specification $\mathcal{B}(C)$: speed compliance, collision rate (with separate early/late targets), event directionality (sigmoid-based toward/away), goal proximity (Gaussian decay), and lingering fraction.

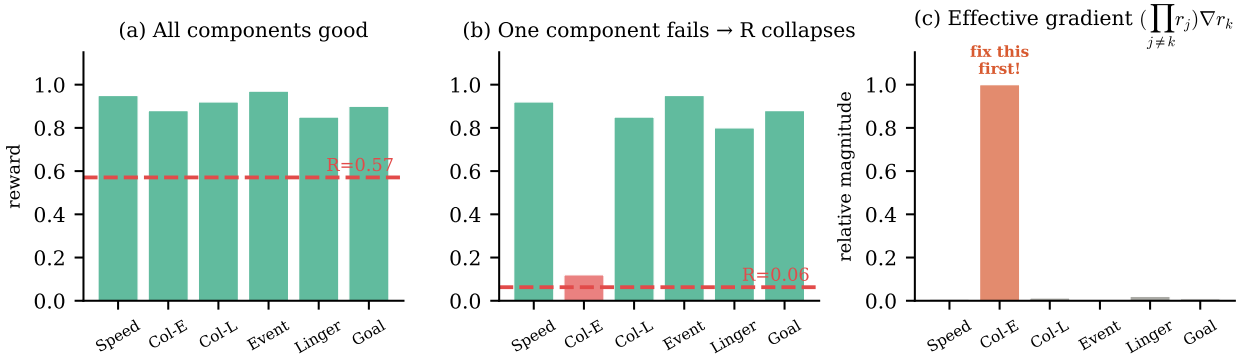

Figure 3: **Multiplicative reward structure.** (a) When all components score well, R is high. (b) If any single component fails (here, collision rate), R collapses even though other components are good. (c) The effective gradient $(\prod_{j \neq k} r_j)\nabla r_k$ concentrates on the worst-performing component, forcing the model to fix it first.

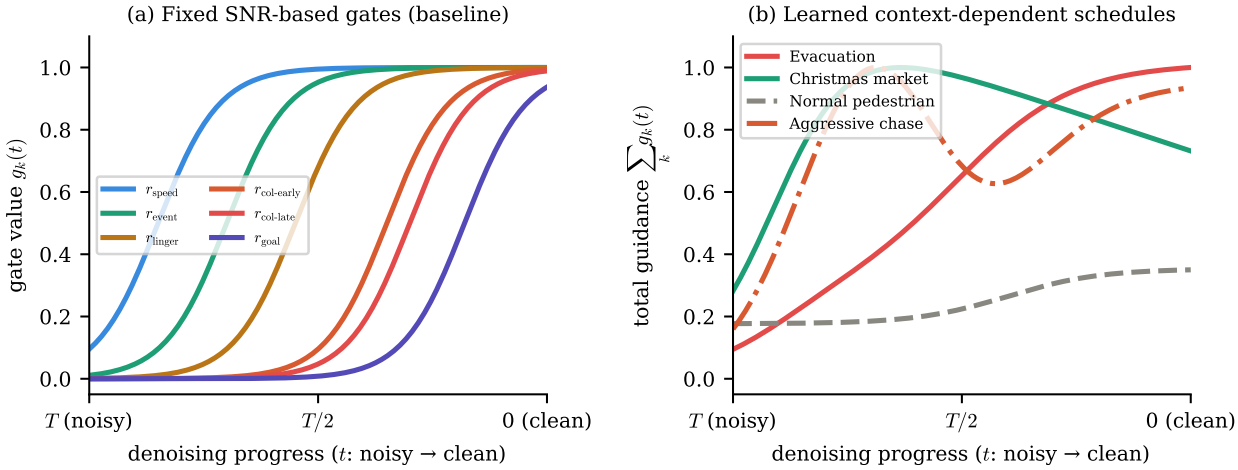

Figure 4: **Guidance schedules.** (a) Fixed SNR-based per-component gates: each sub-reward activates at the noise level where its signal becomes reliable—speed and direction activate early, collision and goal only late. (b) Learned context-dependent schedules (expected): the controller discovers that evacuation contexts need strong late guidance (exit-reaching), market contexts need sustained early guidance (establishing slow speed), and normal contexts need minimal intervention.

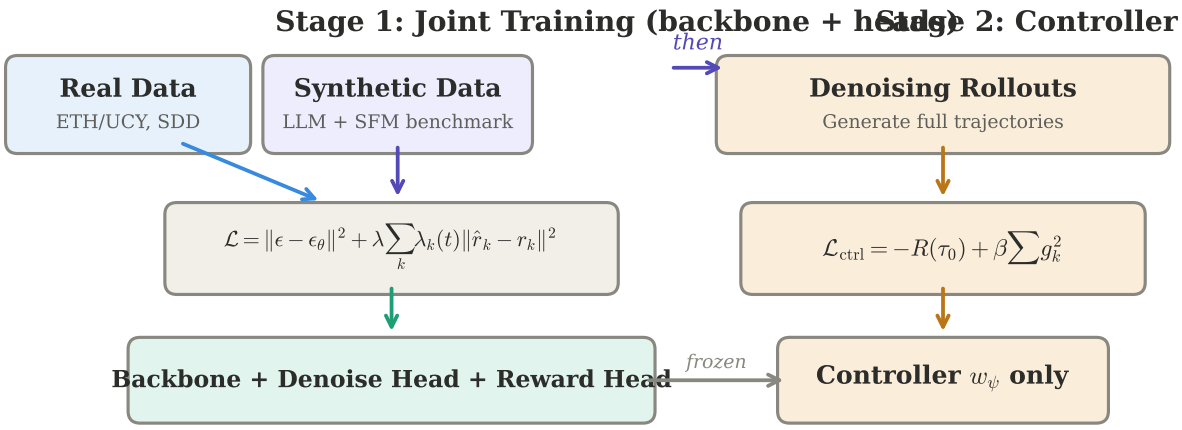

Figure 5: **Two-stage training pipeline.** Stage 1 jointly trains the backbone, denoising head, and reward head on mixed real and synthetic data with the combined denoising + reward loss. Stage 2 freezes all Stage 1 parameters and trains only the lightweight adaptive guidance controller w_ψ via REINFORCE over denoising rollouts on synthetic scenarios.